Cellet Size (μm)	λ	D_h (mm)
212	0.118	1.8
300	0.167	1.8
350	0.194	1.8
500	0.278	1.8

Confinement Ratio (how tightly a particle is confined relative to the scale of the channel):

$\lambda = \frac{\alpha}{D_h}$

Where D_h is diameter/height of channel and α is particle diameter.
Target confinement ratio: $0.1 \leq \lambda \leq 0.3$
Minimum confinement ratio (any smaller and inertia lift force are to week: $\lambda = 0.07$
Maximum confinement ratio (any bigger and risk of particle clogging channel): $\lambda = 0.50$
https://pmc.ncbi.nlm.nih.gov/articles/PMC6906121/

$D_h = \frac{2wh}{(w+h)}$ where w is width of the channel and h is height of the channel
https://semitree.in/tools/hydraulic-diameter/
w = 1.8 mm
h = 1.8 mm
D_h = (2 x 1.8 x 1.8) / (2 x 1.8) = 1.8 mm

Experimentally determine:

- Length of straight section after inlet: initial design: 2mm
- Length of straight section after outlet: initial design: 10mm
- Divider placement at outlet
- The flow rate that gives the clearest seperation

$W_w = P - W_c$
https://www.comsol.com/blogs/how-to-build-a-parameterized-archimedean-spiral-geometry
In this source radial pitch is labeled distance and distance is calculated as 2*pi*b
W_w = Wall gap (thickness between two channels/spirals)
P = Radial Pitch
W_c = Channel width
Chose initial radial pitch of 2.2 mm so each spiral is separated by a 0.4 wall gap
W_w = 2.2 mm - 1.8mm = 0.4 mm

$R(\theta) = r_0 + (p/2\pi)\theta$
https://antennasimulator.com/index.php/knowledge-base/archimedean-spiral/
p = Radial pitch
θ = position (radians)
r_0 = Inner centerline Radius
We want six spirals for θ= 6*(2π) = 12π
R(θ) = 10 + (2.2/2π)(12π) = 23.2mm

Turn	Angular position θ	R(θ)
0	0	10.0
1	2π	12.2
2	4π	14.4
3	6π	16.6
4	8π	18.8
5	10π	21.0
6	12π	23.2

$$Re = \frac{pUD_h}{\mu}$$

$$De = Re\sqrt{\frac{D_h}{2R}}$$

Where Re is the Reynolds number
De is the Dean's number
R is the radius of curvature (centerline curvature)
p is fluid density
U is fluid velocity
μ is dynamic viscosity of the fluid

I will use water a room temperature:
p = 998 kg/m^3
μ =1.002 * 10^{-3} Pa*s

D_h = 1.8 mm
I will initially start with a syringe pump flow rate of 1mL/min
U = Q/A
Q is flow rate and A is crosssection area
A = 0.0018 m * 0.0018 m = 3.24 * 10^{-6} m^2
Q = 1mL/min = 1.666 * 10^{-8} m^3/s
U = 0.00514 m/s

Re = 9.23
De (at inner centerline radius 10mm): 2.77
De (at outer centerline radius 23.2mm): 1.82

In future cure design at 65 C for 40 minute than put in lab oven